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Strong coupling of a microwave photon to an electron on helium

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Abstract

Electrons bound to the surface of superfluid helium have been proposed as a scalable platform for charge- and spin-based quantum computing. Cavity quantum electrodynamics provides a promising method to implement quantum measurement and control of superfluid helium-bound electrons. In this approach, a cavity amplifies the strength of the interaction between the electron and a single photon for coherent exchange of quantum information and qubit readout. This strong coupling regime has been used for quantum measurement in different platforms such as superconducting qubits, atoms and semiconductor quantum dots. Here we demonstrate strong coupling between a microwave photon and the quantized motional state of a single electron on helium using a device comprising a quantum dot and superconducting resonator. Access to this regime provides a basis for developing single-electron spin qubit readout protocols using spin–orbit hybridization techniques that have already been demonstrated in semiconductor quantum dots.

Main

Electrons trapped above the surface of superfluid helium are a compelling system for quantum information science and computing^{1,2,3,4,5}. Electrons confined in nanofabricated traps at the helium surface exhibit quantized motional states³ and share many of the qualities found in quantum dot-based devices in semiconductors, while offering the added advantage of being hosted on a pristine substrate free of local charge traps or nuclear spins. Electrons on helium can be integrated into monolithic architectures to demonstrate high-fidelity shuttling⁶ and precision single-electron control^{7,8}, required for scalable devices. In addition, the electron spin states on helium are predicted to have low spin–orbit interaction and long spin coherence times exceeding 10 s (ref. ⁴), making them an attractive spin qubit candidate.

One particularly promising avenue for spin readout in this system involves coupling the electron motional state to a superconducting microwave resonator using a circuit quantum electrodynamic (cQED) scheme, while inducing spin–charge hybridization with a magnetic field gradient³. However, successful demonstrations in other systems typically leveraged strong coupling to the motional state, where the electron–resonator interaction rate (g) surpasses both the electron decoherence (Γ_2) and resonator linewidth (κ). This ‘strong coupling’ regime has been utilized for quantum measurement across systems including superconducting qubits^{9,10} and atoms^{11,12}. In semiconductor systems, reaching the strong electron motional coupling regime has provided a powerful route for subsequent spin-readout demonstrations^{13,14}.

While the motional states of an electron on helium have been coupled to superconducting microwave cavities^{7,8}, strong charge–photon coupling has remained elusive due to weak interaction between the resonator field and the

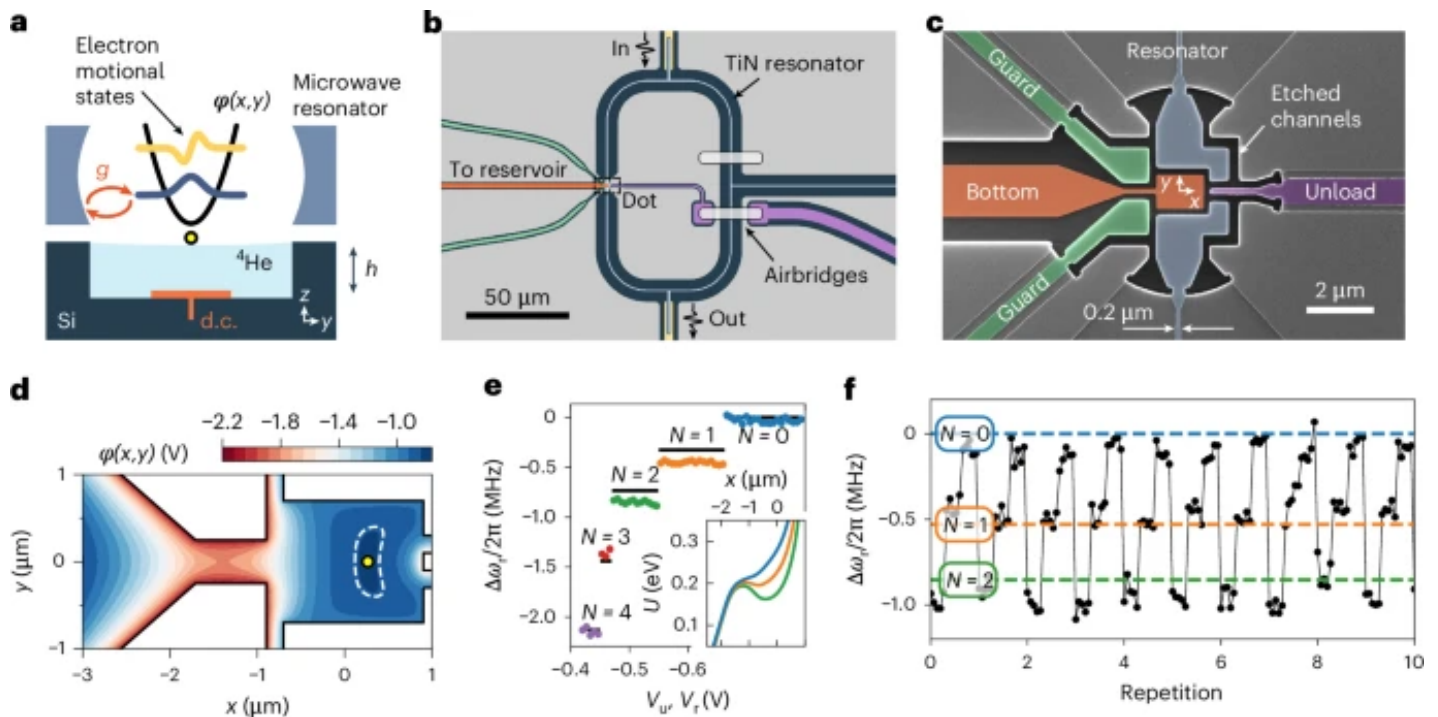
quantized motion of the confined electron⁸. Here, we report experiments that demonstrate strong coupling of the electron motional state in a quantum dot to an on-chip microwave resonator embedded in a microfluidic device. We measure an electron–resonator coupling rate of $g/2\pi = 118$ MHz, which is larger than the resonator linewidth $\kappa/2\pi = 23$ MHz and electron decoherence rate $\Gamma_2/2\pi = 61$ MHz. We demonstrate precise control over individual electrons and their motional states within the quantum dot, in excellent agreement with finite element modelling (FEM) of the device. Furthermore, we show that electrostatic control of the confining potential of the dot allows us to tune the electron charge state frequency and its coherence. These results pave the way towards electron-on-helium-based quantum computing architectures.

cQED device and single-electron loading

An electron trapped above the surface of superfluid helium is bound to its induced image charge in the helium while experiencing a ~1 eV barrier that prevents it from penetrating into the liquid. This provides out-of-plane confinement, but allows lateral control of the electron along the helium surface using electrostatic voltages in silicon-based microfluidic devices¹⁵. The device used for these experiments consists of a single electron on helium, held in a trapping potential that creates a quantum dot and coupled to an on-chip high-impedance coplanar microwave resonator (Fig. [1a–c](#)). Electrons are transported into the dot from an on-chip reservoir ([Methods](#)). In this work, we design a compact quantum dot with dimensions of $1.4 \times 1.4 \mu\text{m}^2$ and implement a high-kinetic inductance resonator with impedance of $Z = 3.8 \text{ k}\Omega$ to boost the coupling energy by a factor of 20 compared with previous realizations of electron on helium quantum dots⁸, resulting in a designed electron–photon interaction rate of $g/2\pi = 110$ MHz (Supplementary Section [1](#)). The quantized motional frequency of the

electron $\omega_e/2\pi$ is tuned by the electrostatic potential in the dot, which is defined by the device geometry and controlled by a combination of voltages V_b , V_r , V_u and V_g , applied to the bottom, resonator, unload and guard electrodes, respectively (Fig. 1c,d). The bottom electrode is below the helium level and is used to transport electrons from the reservoir to the dot (Methods). On the top plane of the device, the guard electrodes provide a potential barrier between the electrons in the dot and those in the reservoir (Fig. 1d). The resonator consists of two titanium nitride (TiN) arms attached to a common bias line for supplying a d.c. voltage, as described in refs. 7,16. The differential mode of the resonator has a frequency $\omega_r/2\pi = 7.162$ GHz and linewidth $\kappa/2\pi = 23$ MHz, which we determine from resonator transmission S_{21} measurements (Methods). The microwave field of the differential mode couples to the electron motion in the dot, allowing the resonator to be used as a sensitive electron detector^{7,8,17}.

Fig. 1: Hybrid electron on the helium cQED device.



a, Schematic of electron motional states coupled to microwave photons in a resonator. Electron is laterally confined above helium surface by trapping potential $\varphi(x, y)$ generated from d.c. voltages on electrodes. **b**, TiN resonator layout with integrated electron dot. Airbridges connect the ground plane and facilitate electrode connections. **c**, False-coloured scanning electron microscope image of the electron trapping region with control electrodes highlighted in colour. **d**, Typical map of $\varphi(x, y)$ in the dot, calculated using FEM, needed to achieve resonance with the motional state and measure features such as those in Fig. [2a](#). The voltages are similar to those presented in Supplementary Table [2](#) and are tuned to promote the quantization axis to be aligned with the y axis, maximizing its coupling to microwave photons. **e**, Steps in the resonator frequency shifts $\Delta\omega_r$ measure the number of electrons in the dot (black lines are modelled values from FEM). Inset: modelled electrostatic potential energy U along the channel for $V_r = V_u = -0.7$ V (blue, $N = 0$), -0.6 V (orange, $N = 1$) and -0.5 V (green, $N = 2$), as required for unloading electrons. **f**, Consistent and repeated loading of $N = 2, 1$ and 0 electrons into and out of the dot. Dashed lines indicate $\Delta\omega_r$ for $N = 2, 1$ and 0 electrons from **e**.

[Source data](#)

We first demonstrate deterministic control of the number of electrons in the dot, enabled by the smooth, defect-free helium surface⁶. The dot is initially loaded with several electrons from the reservoir ([Methods](#)). Electrons are unloaded one at a time as the trapping potential depth is reduced by setting V_r and V_u progressively more negative. Between each unloading event, we return the bias to a probe condition ($V_r = 0.1$ V, $V_u = -0.3$ V) and measure the resonator frequency shift $\Delta\omega_r/2\pi$ at the same electrostatic curvature (Fig. [1e](#)). We infer the number of electrons from the resonator frequency shift $\Delta\omega_r/2\pi$ after each unloading cycle (Supplementary Section [2](#)). For the most negative V_u and V_r , the dot is empty and the resonator returns to its bare resonance ($\Delta\omega_r/2\pi = 0$). To showcase the precise control we have over the electron

number, we repeatedly isolate $N = 2$ electrons from the reservoir and subsequently unload to $N = 1$ and 0 electrons (Fig. [1e](#)). Electrons can be trapped indefinitely because the depth of the electrostatic trap of the quantum dot ($\sim 10^3$ GHz) exceeds both the electron motional state frequency ($\sim 4\text{--}8$ GHz) and the thermal energy (~ 0.2 GHz) by several orders of magnitude, such that quantum tunnelling does not play a role (Supplementary Fig. [5](#)).

Single-electron strong coupling

To characterize the properties of the single-electron cQED system, we load one electron into the dot and measure the resonator frequency shift $\Delta\omega_r/2\pi$ as we change ω_e by independently sweeping the resonator and unload voltages (Fig. [2a](#)) to generate a trapping potential similar to the one in Fig. [1d](#). For a range of voltages, we observe large frequency shifts exceeding several resonator linewidths that appear when the frequency of the electron motion approaches that of the resonator ($\omega_e = \omega_r$), which are confirmed by FEM simulations^{7,18} (Supplementary Section [3](#)). On the upper left-hand side of the resonance contours in Fig. [2a](#), the measured shifts $\Delta\omega_r/2\pi$ are smaller than those observed for other voltage configurations. FEM indicates that this may originate from the population of higher-lying excited states of the electron motion due to relatively small trap anharmonicity for those voltages (Supplementary Fig. [7](#)). Figure [2b](#) presents the resonator transmission $|S_{21}|$ at a fixed resonator bias of $V_r = -2.035$ V (Fig. [2a](#), dashed line), while V_u is swept, revealing two prominent anti-crossings. The transmission signal has been corrected to account for an asymmetric line shape that arises from interference with a background transmission through the resonator¹⁹ (Supplementary Section [4](#)). The two avoided crossings in Fig. [2b](#) correspond to two distinct voltage configurations that produce a resonant interaction between the microwave resonator field and the quantized electron

motional state in the dot (Fig. [2c](#)).

Fig. 2: Single-electron strong coupling and vacuum Rabi splitting.

a, Resonance frequency shift $\Delta\omega_r/2\pi$ from a single electron versus V_r and V_u . The experimental voltages used are presented in Supplementary Table [2](#). **b**, Resonator transmission amplitude $|S_{21}|$ versus V_u at $V_r = -2.035$ V (dashed black line in **a**), revealing two anti-crossings when $\omega_e = \omega_r$. $|S_{21}|$ is normalized to the peak transmission off-resonance and is compensated for interference with direct background transmission across the resonator. **c**,

Simulated first and second excited state wavefunctions for left and right anti-crossings at $V_u = -4.441$ V and $V_u = -4.291$ V using parameters from Supplementary Table 1. **d**, Resonator transmission spectrum $|S_{21}|$ (Supplementary Section 4) for $\omega_e \gg \omega_r$ ($V_u = -4.2$ V, blue trace) and $\omega_e = \omega_r$ ($V_u = -4.291$ V, red trace; arrow in **b**). We extract $g/2\pi = 118 \pm 3$ MHz, $\Gamma_2/2\pi = 75 \pm 5$ MHz and $\omega_e/2\pi = 7.169 \pm 0.004$ GHz.

[Source data](#)

In the strong coupling regime, the coherent hybridization of electron and resonator quantum states produces two clearly distinguishable modes in the resonator transmission spectrum, also referred to as vacuum Rabi splitting²⁰. Figure 2d shows the magnitude of the complex transmission $|S_{21}|$ as the resonator probe frequency is tuned for two values of V_u . At $V_u = -4.2$ V, $\omega_e \gg \omega_r$, and the spectrum is close to that of a bare resonator (blue trace). However, at $V_u = -4.291$ V, $\omega_e = \omega_r$, and two distinct peaks emerge, separated by the vacuum Rabi splitting $2g/2\pi$ (red trace), which unambiguously confirms that the electron is strongly coupled to the resonator. We fit the spectra in Fig. 2d to input–output theory of the coupled electron–resonator system (Supplementary Section 4) and extract a coupling strength of $g/2\pi = 118 \pm 3$ MHz and an electron decoherence rate of $\Gamma_2/2\pi = 75 \pm 5$ MHz, corresponding to a single photon cooperativity $\mathcal{C} = 4g^2/\kappa\varGamma \approx 32$. The experimentally extracted value of g shows good agreement with the designed value, indicating the accuracy of FEM simulations in capturing the electric field distribution within the quantum dot. This contrasts with semiconductor quantum dot systems, where detailed knowledge of the electric field profile is often limited due to fabrication-induced variability and nearby charge defects, making it challenging to accurately determine the electron's position^{21,22}.

Electron charge state spectroscopy

We next directly probe the electron spectrum by performing two-tone spectroscopy²³ using the pulse sequence in Fig. 3a. We begin by shaping the electrostatic trapping potential such that the electron motional state is far detuned from the bare resonator frequency by ~1 GHz. We then apply a variable-frequency drive tone $\omega_d/2\pi$ to the resonator for a fixed time to excite the electron, immediately followed by a readout tone at the resonator centre frequency $\omega_r/2\pi$. As shown in Fig. 3a, the dispersive interaction between the electron and resonator leads to a decrease of the transmitted readout signal $\Delta|S_{21}|$ when $\omega_d/2\pi$ is resonant with the electron motional state transition. Here, we use the definition $\Delta|S_{21}|(\omega_d) = |S_{21}|(\omega_d) - |S_{21}|(\omega_{\text{off}})$, where ω_{off} is a drive frequency far off resonance such that $|\omega_{\text{off}} - \omega_e| \geq 1 \text{ GHz}$. By fitting the resulting signal to a Lorentzian, we extract the electron motional frequency $\omega_e/2\pi$ and two-tone linewidth $\gamma/2\pi$, which is in reasonable agreement with the values determined from resonator spectroscopy.

Fig. 3: Electron charge state spectroscopy.

a, Spectroscopy drive pulse (frequency $\omega_d/2\pi$, duration 5 μs) followed by resonant readout pulse (frequency ω_r , duration 1.5 μs) applied to the resonator input port. A dip in readout transmission occurs at the electron frequency $\omega_e/2\pi = 8.66$ GHz with a half width at half maximum $\gamma/2\pi = 102 \pm 2$ MHz (ref. [23](#)). **b**, Electron motional state spectrum (blue colour map) versus V_u , showing two voltages where $\omega_e \approx \omega_r$, which agree with the avoided crossings visible in the resonator spectroscopy (red colour map and dashed red curve). The simulated electron frequency based on $\Delta\omega_r$ (dotted grey curve) agrees with the measured ω_e (Supplementary Section [5](#)). Areas where $\omega_e/2\pi \approx 7.162$ GHz are omitted due to fast leakage of the drive tone through the resonator.

[Source data](#)

To demonstrate control over the quantized electron motional states, we extract the electron frequency from two-tone spectroscopy measurements as we tune ω_e with V_u . In Fig. [3b](#), we highlight the agreement between the electron frequency determined from resonator spectroscopy (dashed red curve), two-tone spectroscopy (blue data) and FEM (dotted grey curve) as we tune the trapping potential. In particular, the

electron frequency we extract from resonator spectroscopy by using the dispersive approximation $\omega_e = \omega_r + g^2/\Delta\omega_r$ (Supplementary Section 5), closely follows the values determined from two-tone spectroscopy for an electron–photon coupling strength of $g/2\pi = 110$ MHz over a broad range of voltages, implying that the coupling does not depend strongly on V_u . This value of g is also in agreement with the independently determined value of the coupling strength from the vacuum Rabi splitting. The non-monotonic dependence of $\omega_e/2\pi$ on V_u arises from the fact that the minimum of the trapping potential moves off-centre while sweeping V_u (Fig. 2c). Both an analytical model and FEM simulations suggest that our observation that the electron frequency never falls below 5.5 GHz can be explained by an in-plane electric field at the location of the electron, potentially arising from stray charges in the vicinity of the dot (Supplementary Section 6).

Electron state decay and dephasing

Two-tone spectroscopy of the electron motional state spectrum provides a stepping stone towards time-domain measurements, for example, to probe the electron's energy relaxation time. The electron state decoherence contains contributions from both energy relaxation and pure dephasing ($\Gamma_2 = \gamma_1/2 + \gamma_\phi$). To determine the dominant contribution to the total linewidth, we perform time-domain measurements to extract the electron energy relaxation time, $T_1 = 1/\gamma_1$, as shown in Fig. 4a. We apply a 20- μ s pulse at frequency $\omega_e/2\pi$, followed by a variable length delay τ and subsequent readout pulse at $\omega_r/2\pi$. As τ increases, the transmitted readout signal decays exponentially with a characteristic time constant T_1 , which we extract from fitting. The data in Fig. 4a are taken for three different electron frequencies corresponding to differing values of the electron–resonator frequency detuning (blue, green and orange). At large electron–resonator detuning, we find a maximum

$T_1 \approx 0.76 \mu\text{s}$, while at smaller detuning values, energy relaxation is enhanced, probably from Purcell decay into the resonator (Supplementary Section [7](#)). In addition, as shown in Fig. [4b](#), we find that γ_1 is temperature independent, consistent with spontaneous emission. Importantly, these measurements show that γ_1 is two orders of magnitude smaller than Γ_2 , indicating that the electron decoherence is primarily limited by pure dephasing, not energy relaxation.

Fig. 4: Energy decay and dephasing of the electron motional state.

a, T_1 measurements taken for three different V_u for $\omega_e < \omega_r$ (orange), $\omega_e > \omega_r$ (blue) and $\omega_e \approx \omega_r$ (green). The decay rate increases due to Purcell loss through the resonator when the electron–resonator detuning is small (green). Inset: the electron frequency dependence from Fig. 2b, with arrows indicating the electron frequency at each T_1 measurement. **b**, Electron decay rate γ_1 versus temperature, consistent with spontaneous emission. **c**, Decoherence rate Γ_2 decreases when negative voltage offsets ΔV are applied to the dot electrodes. The dashed line indicates $g/2\pi = 110$ MHz. Inset: offset ΔV lifts the potential energy relative to the ground potential $\varphi(y) = 0$. Dot electrode voltages used for each data point are provided in Supplementary Table 3. While more negative offsets result in a reduction in Γ_2 , there is an experimental limit beyond which the electron is no longer confined in the dot. However, the voltage offsets can potentially be made more negative in future devices optimized for improved lateral confinement. **d**, Decoherence rates Γ_2 measured at temperatures between 7 mK and 450 mK. The dashed line with a slope of 0.5 is plotted on the log–log scale as a guide to the eye. Data in **b–d** are presented as best-fit values, with error bars representing fitting uncertainty.

[Source data](#)

Previous attempts at reaching the strong coupling regime with electrons on helium suggested that low-frequency classical helium surface fluctuations could be a dominant decoherence mechanism⁸. These surface fluctuations, arising from mechanical vibrations in the system, can in principle modulate the electron's out-of-plane position, leading to variations in the in-plane trapping potential. This can alter the electron frequency and potentially cause dephasing. However, we observed no substantial change in the two-tone linewidth γ when the level of mechanical vibration of our cryostat was increased by an order of magnitude by disabling the active vibration cancellation system (Supplementary Section 8).

To investigate what limits Γ_2 , we study the effect of modifying the trapping potential on the electron decoherence rate by applying a negative offset ΔV to the trapping electrodes relative to the nearby ground electrode (Fig. 4c). Notably, a transition from the weak to strong coupling regime is observed as the offset is made progressively negative, with the lowest measured decoherence rate reaching $\Gamma_2/2\pi = 61$ MHz. This transition is unambiguously confirmed by the appearance of a vacuum Rabi splitting in the measured resonator spectroscopy for voltage configurations with offset $\Delta V < -1$ V (Supplementary Section 9). The observed decrease in Γ_2 may be attributed to a reduced electron dephasing as the electric field pressing the electron against the helium surface becomes weaker. This behaviour is expected for electrons interacting with the bosonic field of quantized surface capillary waves (ripples)²⁴. Alternately, the voltage offset applied to the electrodes may change the spatial distribution of nearby stray electrons that are potentially introduced during the initial electron emission process and that couple to the trapped electron. Applying more negative ΔV could repel these fluctuating stray charges (Fig. 4c), thereby reducing dephasing from charge noise, a ubiquitous decoherence mechanism for solid-state qubits^{25,26}. To distinguish between these two effects, future devices and packaging can be designed so that electron emission occurs in an area far removed from the dot to mitigate the effects of stray charges.

We further demonstrate that the decoherence rate has a strong temperature dependence, as shown in Fig. 4d, changing almost tenfold from 7 mK to 450 mK (Supplementary Section 9). The value of Γ_2 closely follows a power-law dependence T^β with $\beta \approx 0.5$ and shows no sign of saturation at the lowest temperatures. This behaviour is consistent with the coupling of the electron state to a thermally activated bath having low-frequency excitations^{24,27}, which could originate from either the ripplonic field of the helium surface or the fluctuation of stray charges in

the vicinity of the dot.

Discussion and outlook

In conclusion, our result of strong coupling between the electron motional state on helium and the microwave resonator photons opens access to the investigation of a range of new light–matter phenomena with a single fundamental particle. Future material and design improvements²⁸ may further enhance the coupling rate to enable coherent control of electron on helium charge qubits and access to exotic ultrastrong coupling regimes of circuit quantum optics²⁹ with individual electrons or electron ensembles^{30,31}. Importantly, our demonstration of strong coupling is a vital ingredient for interfacing an electron spin on helium to a microwave photon using well-established cQED methods similar to those used in semiconductor quantum dots^{13,14}. These techniques typically employ a micromagnet structure that can readily be integrated into our existing device geometry to induce spin–orbit hybridization in the presence of a local magnetic field gradient, therefore enabling spin detection via the motional state³. Notably, previous demonstrations with similar charge qubit decoherence and coupling rates have achieved spin readout via charge state hybridization¹³, placing microsecond spin qubit readout on helium within reach (Supplementary Section [10](#)).

Methods

Device fabrication

The single-electron devices are fabricated on undoped high-resistivity Si wafers (>10 k Ω cm) with (100) orientation. Immediately after a standard solvent cleaning of the wafer, a 12-nm-thick TiN film is grown by Plasma-Enhanced ALD

(Ultratech/Cambridge Fiji G2) at a temperature of 270 °C and 150 cycles. This superconducting film is used to fabricate the resonator, electrodes and ground plane in our devices.

After the TiN film growth, the geometry of the electron reservoir, the microchannel and the electron dot is defined using a combination of e-beam lithography and dry etching. The e-beam lithography is performed with a Raith EBPG5000+ to form a resist etch mask on the TiN film surface. Subsequently, we etch Si/TiN at once using fluorine chemistry in an ICP plasma etcher. The dry etching recipe uses a gas mixture of SF₆ 20 sccm:C₄F₈ 30 sccm, ICP power 600 W and radio frequency bias power 60 W at 10 °C and 10 mTorr pressure. To remove the remaining resist after etching, the sample is soaked in AZ *N*-methyl-2-pyrrolidone (NMP) rinse at 80 °C overnight, followed by 30 min of sonication with a fresh AZ NMP rinse at 80 °C and an isopropanol (IPA) rinse at the end. This process forms ~0.78-μm-deep TiN-covered Si trenches (Supplementary Fig. [1a](#)), where liquid helium is filled up by capillary action in the experiment.

The next patterning step involves fabricating the resonator, guard, and unload electrodes by dry etching TiN using a resist etch mask, as shown in Supplementary Fig. [1b](#). The etch mask is first patterned on the sample using e-beam lithography (Raith EBPG5000+), and then the area of TiN not protected by the mask is etched away using the same recipe and the same tool as above. The total etching time of TiN is extended by ~50% to ensure there is no TiN residue, leading to additional Si etching by ~90 nm. After etching TiN, the resist mask is first partially removed by an oxygen plasma using a downstream plasma asher YES-CV200 RFS, followed by putting the sample in an AZ NMP rinse at 80 °C overnight. The sample is sonicated with fresh NMP at 80 °C for 1.5 h and rinsed with IPA at the end.

To add additional electrodes, we perform another patterning process using a lift-off method. First, an e-beam resist pattern is formed with another round of e-beam lithography, followed by an 80-nm-thick Nb layer deposition using a Plassys MEB550S Electron Beam Evaporator. For resist removal after Nb deposition, the sample is soaked in AZ NMP rinse at 80 °C overnight, and then sonicated with a fresh AZ NMP rinse at 80 °C for 20 min. The remaining NMP is cleaned away by an IPA rinse. As a result, the bottom electrode in the dot region is made of Nb situated at the bottom of the ~0.78- μm -deep TiN-covered Si trench (light-blue electrode in Supplementary Fig. [1g](#)). The resonator feedline and d.c. bias lines on the TiN ground plane are formed from Nb placed in the ~100-nm-deep TiN etched region (light-blue electrodes in Supplementary Fig. [1c](#)). The guard and unload electrodes (TiN) are electrically connected to their own Nb d.c. bias lines through the Al airbridges, the fabrication of which is described below.

Using optical lithography, we first form the scaffold to support the airbridges with 3- μm -thick positive photoresist. The resist is then reflowed at 140 °C for better mechanical stability. Next, 300-nm-thick aluminium is deposited under high vacuum using a Plassys MEB550S Electron Beam Evaporator to build the metal layer of the airbridge. Before the aluminium deposition, in situ ion milling is performed to remove the native oxide on the TiN and Nb surface for better ohmic contact with the aluminium. Through another round of optical lithography, the bridge pattern is defined with the same thick photoresist, and the aluminium not used for the airbridge is removed by wet etching with Transene Aluminum Etchant Type A. Before aluminium wet etching, the resist is baked at 115 °C to improve adhesion to the aluminium. After wet etching, the resist is removed by a combination of dry and wet resist strip processes. During the dry strip process, most of the cross-linked resist formed during the ion milling step before the aluminium deposition is removed by

oxygen plasma. The sample is then soaked in AZ NMP rinse to remove all remaining resist, followed by rinsing with IPA. Supplementary Fig. [1d](#) shows the Al airbridge (red) used to connect the TiN unload electrode (blue) to the Nb d.c. bias line (light blue). The TiN guard electrodes are also connected to their Nb d.c. bias lines in the same way, which is not shown in Supplementary Fig. [1](#). In addition to providing electrode connections, the Al airbridges are also placed over most signal lines to connect the two TiN ground planes on either side of the lines, thereby suppressing slotline modes.

Sample cell and helium filling

The fabricated chip of size 2 mm × 7 mm is wirebonded to the host PCB, which is hermetically sealed within the sample cell, as schematically illustrated in Supplementary Fig. [2](#). An additional volume beneath the chip serves as a bulk liquid helium reservoir. The top section of the cell is connected to a large 150 cc reservoir used to store helium gas before cooldown. A small tungsten filament positioned several millimetres above the chip serves as the source of electrons.

The ^4He is isotopically enriched and sourced from Lancaster Helium. We pressurize the sample cell at room temperature to 0.9 bar, corresponding to a bulk liquid helium volume of approximately 0.18 cm³. A liquid nitrogen cold trap between the gas cylinder and the experimental cell is used to filter impurities such as water and nitrogen. Immediately before cooling down, we seal the experimental cell.

Below the superfluid transition temperature, liquid helium fills the on-chip microchannels via capillary action, with a radius of curvature set by the distance from the chip to the bulk helium reservoir H (Supplementary Fig. [2](#)). From the sample cell geometry, we estimate $H \approx 3$ mm, which results in a depression of the helium surface

at the centre of the dot of approximately $\left(\frac{\rho_{\text{He}} G H_{\text{ch}}^2}{8 \sigma_{\text{He}}}\right) \approx 2 \text{ nm}$. Here, ρ_{He} and σ_{He} are the density and surface tension of superfluid helium, respectively, $w_{\text{ch}} = 1.4 \text{ }\mu\text{m}$ is the width of the electron dot and $G = 9.81 \text{ m s}^{-2}$ is the Earth's gravitational acceleration.

Cryogenic set-up

All measurements were performed in a Bluefors LD400 dilution refrigerator with a base temperature of 7 mK. The experimental configuration for electron spectroscopy measurements of Figs. 3 and 4 is shown in Supplementary Fig. 3.

Electrode d.c. voltages are supplied by a QDevil QDAC II that connects to the Bluefors twisted pair loom. We filter d.c. lines at the mixing chamber stage, with a stack consisting of a QDevil RC filter, a custom-built filter board containing several π -filters and, lastly, a filter box filled with infrared absorber to filter frequencies above 1 GHz. We measure a cut-off frequency for this filter stack of approximately 2 kHz, which still allows fast voltage sweeps if needed. The outputs of the filter stack go directly to the hermetic cell, except for one of the guard electrodes and the bottom electrode, which instead pass through a bias tee. The radio frequency port of both bias tees is not used in this experiment.

Electron reservoir

Electrons are emitted from the tungsten filament located inside the hermetic sample cell. One terminal is connected to a Keysight 33220A arbitrary waveform generator (configured in single pulse mode), while the other end is grounded. We typically observe electron emission by pulsing the filament for a duration of 100 ms and amplitude of -2.4 V , while the cell is thermalized at a temperature of 0.65 K. Electrons

are then captured by applying a positive voltage (typically 1.5 V) to the centre pin of a niobium $\lambda/2$ -resonator ($\omega_r/2\pi = 6.527$ GHz), located on the bottom of a 0.7- μm -deep microchannel. This results in a dispersive resonance frequency shift of several megahertz signalling the presence of electrons on top of the reservoir resonator. Because the microchannel connects to the dot shown in Fig. [1b](#), electrons can now be transferred from the reservoir resonator to the dot with the appropriate voltages.

To transfer electrons into the dot, we bias the bottom electrode at 0.4 V and the guard electrodes close to 0.0 V, and decrease the voltage on the reservoir resonator until a negative frequency shift is observed on the TiN resonator. During experiments, the reservoir is kept at 2.0 V such that any stray electrons close to the dot are drained to the reservoir resonator.

Electron readout and spectroscopy pulses are generated by a Quantum Machines OPX and Octave unit. Both readout and spectroscopy pulses are combined at the TiN readout resonator input line, are attenuated by approximately -73 dB and then filtered (Quantum Microwave IR filter and K&L 6L250 low-pass filter). The transmission is filtered and amplified by a Low Noise Factory (LNF) high-electron mobility transistor amplifier, and further amplified at room temperature before the signal is processed by the Octave.

We probe the electron reservoir resonator with a vector network analyser (VNA, Keysight E5071C) and measure the resonator in reflection (S_{11}) using a double junction circulator (LNF CICIC 4_8A) mounted at the mixing chamber. Except for the final high-pass filter, the return chain for the reservoir resonator is identical to the return chain of the TiN resonator.

FEM of trapping potential

The behaviour of both the many-electron system and a single electron within the dot region is entirely governed by the local electrostatic potential at the helium surface. This potential is controlled via the voltages V_i applied to the electrodes defining the dot area and is expressed as $\varphi(x, y) = \sum_i \alpha_i(x, y) V_i$, where $\alpha_i(x, y)$ represents the relative contribution of the i th electrode to the overall potential. We employ FEM to compute $\alpha_i(x, y)$ by setting each i th electrode to 1 V while grounding all other electrodes at 0 V, repeating this procedure for every electrode in the system. The resulting potential is then used to determine the many-electron ground-state configuration and their vibrational frequencies. The single-electron excitation spectrum is obtained by numerically solving the two-dimensional Schrodinger equation for an electron in a trapping potential $\varphi(x, y)$, from which we extract the electron frequencies ω_e and anharmonicities α_e . We also compute the microwave field distribution produced by the resonator pins and use it to estimate $\langle \ell^{-1} \rangle = \langle \partial \alpha / \partial y \rangle$. Here, α^- is obtained by setting upper and lower arms of the resonator to +1/2 V and -1/2 V, respectively, and all other electrodes are kept at zero potential. This estimate is then used to determine the electron–photon coupling strength.

Data availability

The datasets generated and analysed during the current study are available via Zenodo at <https://doi.org/10.5281/zenodo.20086616> (ref. ³²). [Source data](#) are provided with this paper.

Code availability

The codes used to analyse the datasets are available via Zenodo at <https://doi.org/10.5281/zenodo.20086616> (ref. ³²).

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Contributions

G.K. and E.O.G. performed the experiments with assistance from N.R.B., D.G.R. and K.E.C. The data were analysed by G.K. and E.G. N.R.B. conceived of the device design. N.R.B. and M.S. designed and simulated the device, which was fabricated by H.B. Theoretical calculations to support the experimental work were primarily performed by G.K. and M.S. E.O.G., G.K., N.R.B and J.P. prepared the paper with input from all authors. J.P., S.A.L. and D.G.R. supervised and assisted in the project and provided guidance.

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Ethics declarations

Competing interests

The authors declare no competing interests.

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